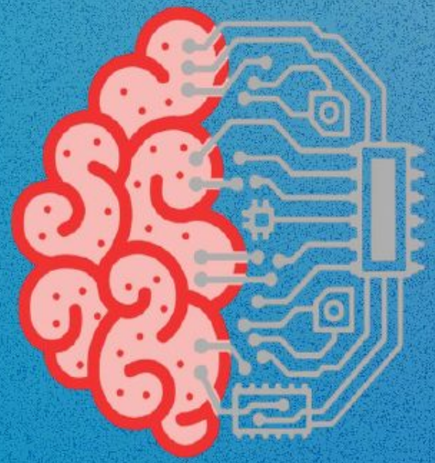
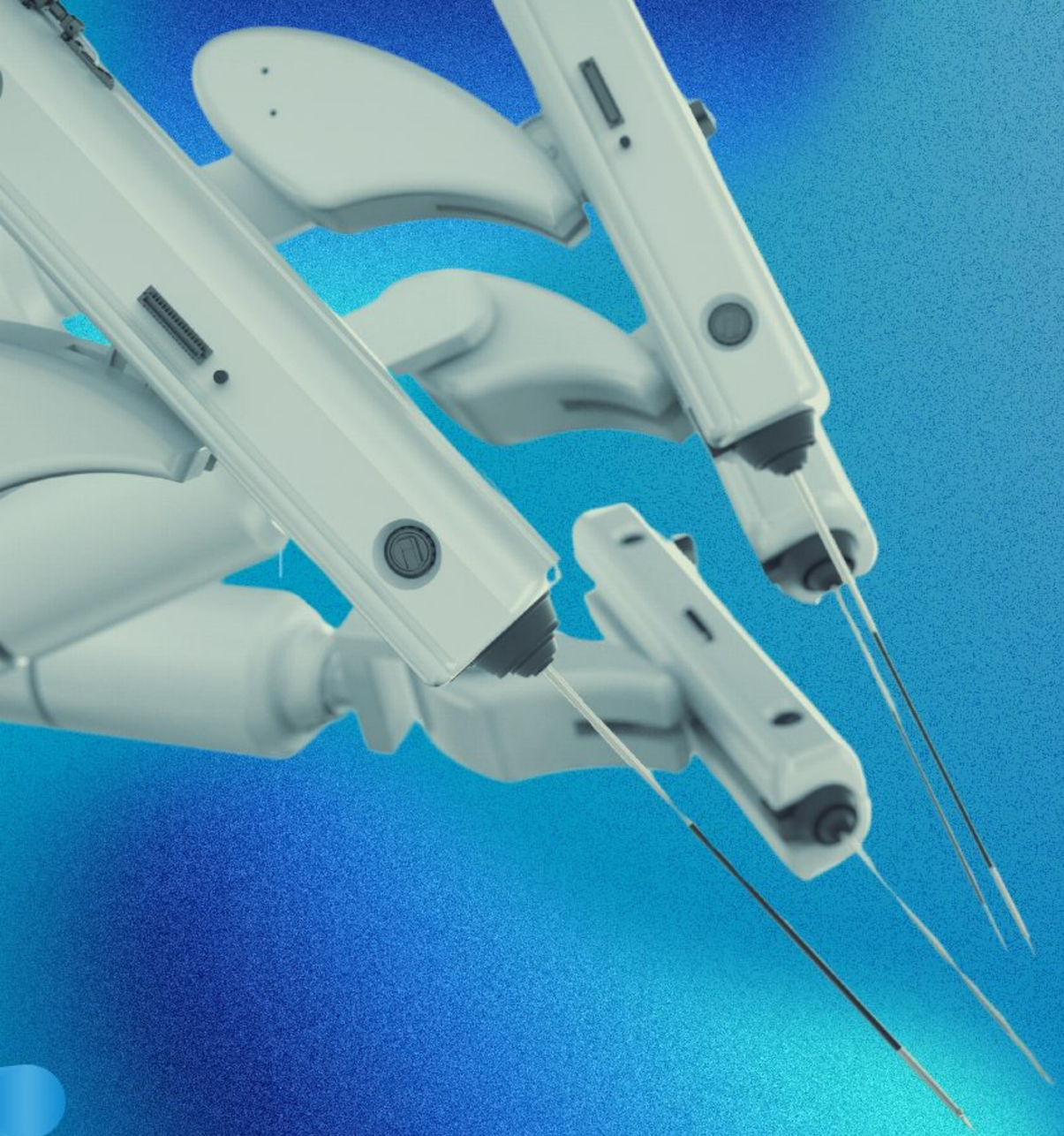
AI in Health and Medicine:

Lessons for Higher Education

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ARTIFICIAL INTELLIGENCE IN MEDICINE AND HEALTHCARE ASSOCIATION





Agenda

Why it matters

Ethical issues

Legal issues

What it means for us

Discussion

Poll

Who has used AI for coursework, research, or studying this semester?

Who knows Rutgers' AI policy?





Framing Idea

- AI isn't simply good or bad. It depends on the case and which values are at stake (Savulescu et al., 2024)
- Someone always decides what goals and values a system serves, so ethics can't be avoided
- AI is a human tool, and oversight has to come with its design and use



Why This Matters

AI is already everywhere, and training lags



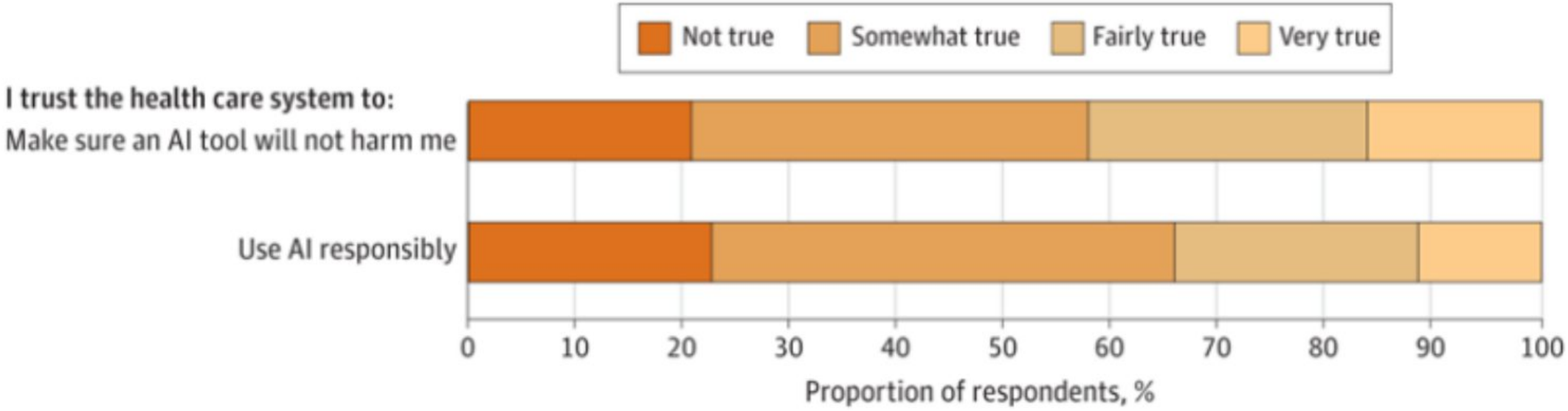
- 55.7% of UK doctors surveyed had used AI programs, but only 7.1% reported formal AI training (Warrington & Holm, 2024)
- Most rated themselves only slightly (44.3%) or moderately (31.4%) knowledgeable
- 1,600+ AI-enabled medical devices authorized in the U.S. as of Sept 2026 (FDA)
- Takeaway: widespread use, little guidance, which is the same situation as students and faculty

The Trust Gap

- 65.8% of U.S. adults had low trust in their health system to use AI responsibly; 57.7% doubted it would protect them from harm (Nong & Platt, JAMA Netw Open, 2025, n=2,039)
- Those with high general trust were 4x as likely to trust its AI use; people who experienced discrimination were less likely to
- Neither AI knowledge nor health literacy predicted trust
- Wolters Kluwer 2026: the trust gap is widening even as use rises

The Trust Gap

Figure. Trust in Health Care Systems to Use Artificial Intelligence (AI) Responsibly



Demographic Discrepancies

	Extremely/ Very	Somewhat	Not too/ Not at all
U.S. adults	72	16	7
Men	68	18	8
Women	74	14	5
White	75	15	6
Hispanic	66	16	9
Black	67	17	6
Asian*	62	23	8
Ages 18-29	68	14	10
30-49	69	16	7
50-64	73	18	5
65+	76	14	5
Postgrad	70	17	8
College grad	75	16	6
Some college	75	15	5
HS or less	67	16	8

Ethical Considerations

Reliability and Accuracy

- NIH study: GPT-4V answered 207 medical image quiz questions and beat physicians working without resources, but physicians with resources did better, especially on hard questions
- It often described images wrongly or explained its reasoning wrongly even when the final answer was correct
- 74% of clinicians cite AI inventing false info (like fake studies) as a major concern (Wolters Kluwer)
- AI isn't held to drug-level testing. An IVF embryo-selection AI was introduced without randomized trials against embryologists
- Limitation to note: the NIH study was small



Overreliance and Deskilling

- Ratti et al.: physician deskilling is an emerging concern, because heavy AI dependence can weaken independent judgment
- 74% of clinicians worry that overreliance weakens their ability to catch errors
- Yet 77% of clinicians double-check AI against trusted sources like PubMed or UpToDate
- Higher-ed parallel: critical thinking and learning to evaluate sources



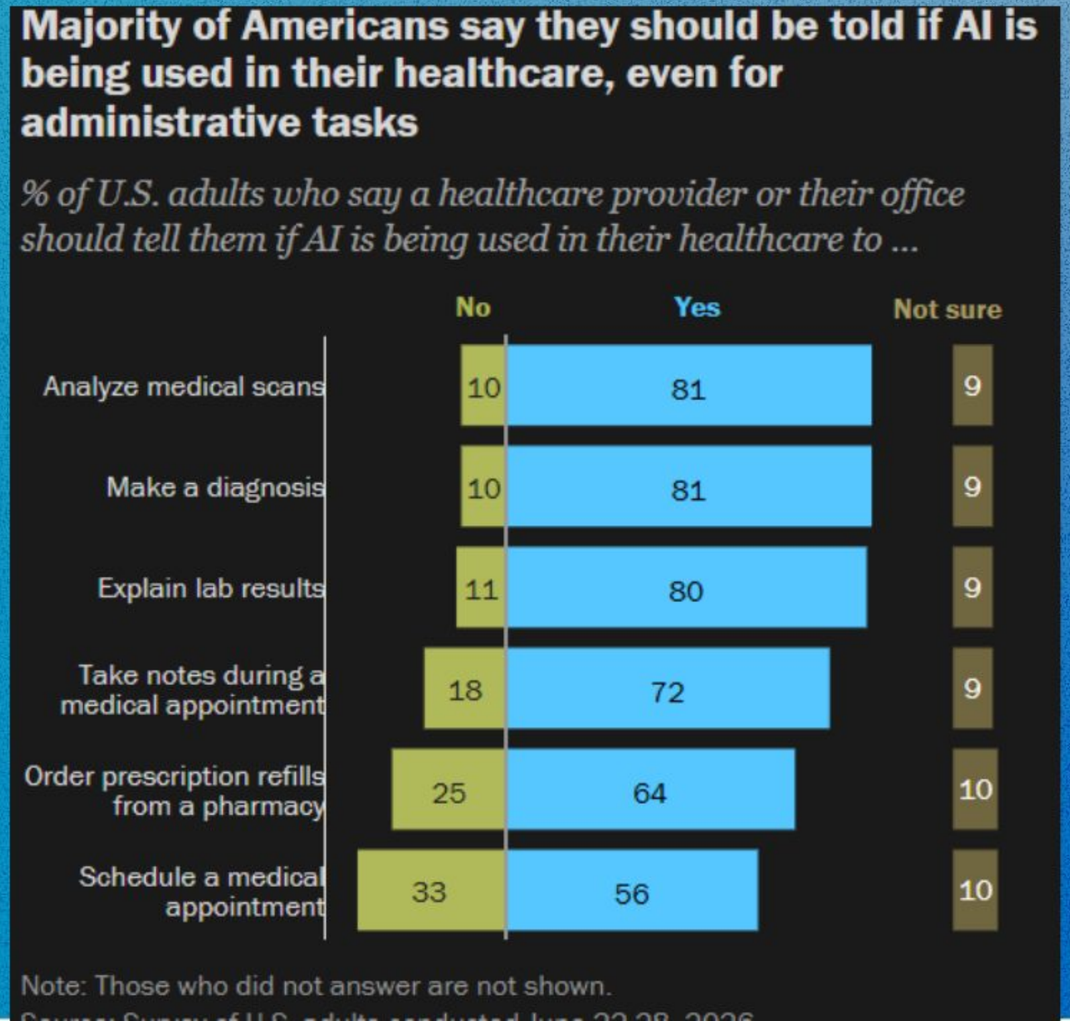
Privacy and Data

- AI needs big data, which raises confidentiality, consent, and unauthorized-sharing risks (WHO)
- Top concerns among UK doctors: patient confidentiality, companies using patient data, bias, legal responsibility
- De-identified data can still be re-identified (Savulescu)
- Proposed fixes: broad, meta, and dynamic consent, plus independent data authorities
- Ratti trade-off example: more data from children could improve dosing research but raises privacy risk

Transparency, autonomy, and informed consent

- Pew (June 2026, n=3,488): 72% say it's extremely or very important that providers tell them if AI is used
- 46% aren't sure whether AI was used in their care; 63% want more say
- Majority want disclosure even for administrative tasks (56% for scheduling, 81% for scans and diagnoses)
- Machine paternalism: IBM Watson for Oncology ranked treatments by length of life, not quality of life, which built in programmers' values
- Higher-ed parallel: should students be told when AI is used in grading, advising, or admissions?

Transparency, autonomy, and informed consent

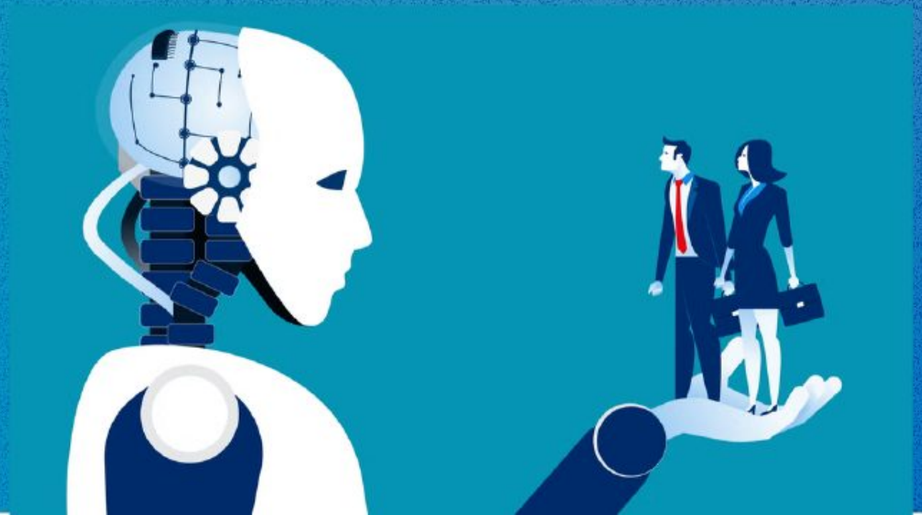


Equity and Access

- Pew: disclosure matters to 75% of White, 67% of Black, 66% of Hispanic, and 62% of Asian adults
- 25% of postgraduates vs 11% of those with high school or less said their providers had used AI
- Ratti: cost raises concerns about whether poorer hospitals and communities get the same access
- WHO: AI could help rural communities but shouldn't replace essential services
- Higher-ed parallel: who has access to premium AI tools, and who gets left behind?

Trade-offs and conflicts of interest

- Ethical improvements involve trade-offs (Ratti)
- 72% of clinicians and 61% of patients worry that ad-sponsored AI could bias decisions (Wolters Kluwer)
- WHO: without regulation, companies' financial interests could outrank patient rights
- Higher-ed parallel: vendor-driven AI adoption on campuses



Legal Considerations



- A physician is liable ONLY if care falls below the standard, whatever the AI recommended (Price, Gerke & Cohen, JAMA, 2019)
- In a juror study, physicians who accepted standard-care AI advice faced less liability than those who rejected it (Tobia et al., J Nucl Med, 2021)
- The physician is accountable for a tool they didn't design and can't fully check -> incentive is to follow the standard rather than think independently

CONCLUSION

Safest use for liability is AI as a confirmatory tool

Legal Considerations

Physicians' regulatory priorities (AMA, 2026):

1. **Clear liability rules**
2. **Post-market monitoring**
3. **FDA oversight of AI devices**

- **88% say validated safety and efficacy matters for adoption, and 86% say the same of data privacy ama-assn**
- **Privacy concern as 71% for tools outside the institution vs. 42% for institution-provided tools ama-assn**
- **Safety, privacy, and accountability are institutional responsibilities, not individual ones**
- **Physicians ranked patient consent and disclosure last, yet 72% of the public wants to be told when AI is used (Pew)**

What it means for us

88% of doctors worry about skill loss (AMA, 2026)

70% of doctors worry about skill loss in med students & residents (AMA, 2026)

92% want more AI training, 27% had none, 11% trained got a lot (AMA, 2026)

- “Human in the loop” only works if the human can still catch errors
- Use is outpacing training, so we are still building the skills future oversight depends on
- Learn to do it first, then use AI to extend it

What it means for Higher Education



Healthcare Issue	Higher Education Parallel
Algorithmic Bias	Bias in admissions, grading, or proctoring tools
Hallucinations and Errors	Fake citations in student work
Deskilling	Weakened critical thinking
Patient Privacy	Student data privacy
Disclosure and consent	Telling students when AI is used on them
Unequal access	The digital divide in AI tools
Accountability	Who is responsible for AI errors?
Lack of Training	Few students or faculty trained in AI

AI for Rutgers Students

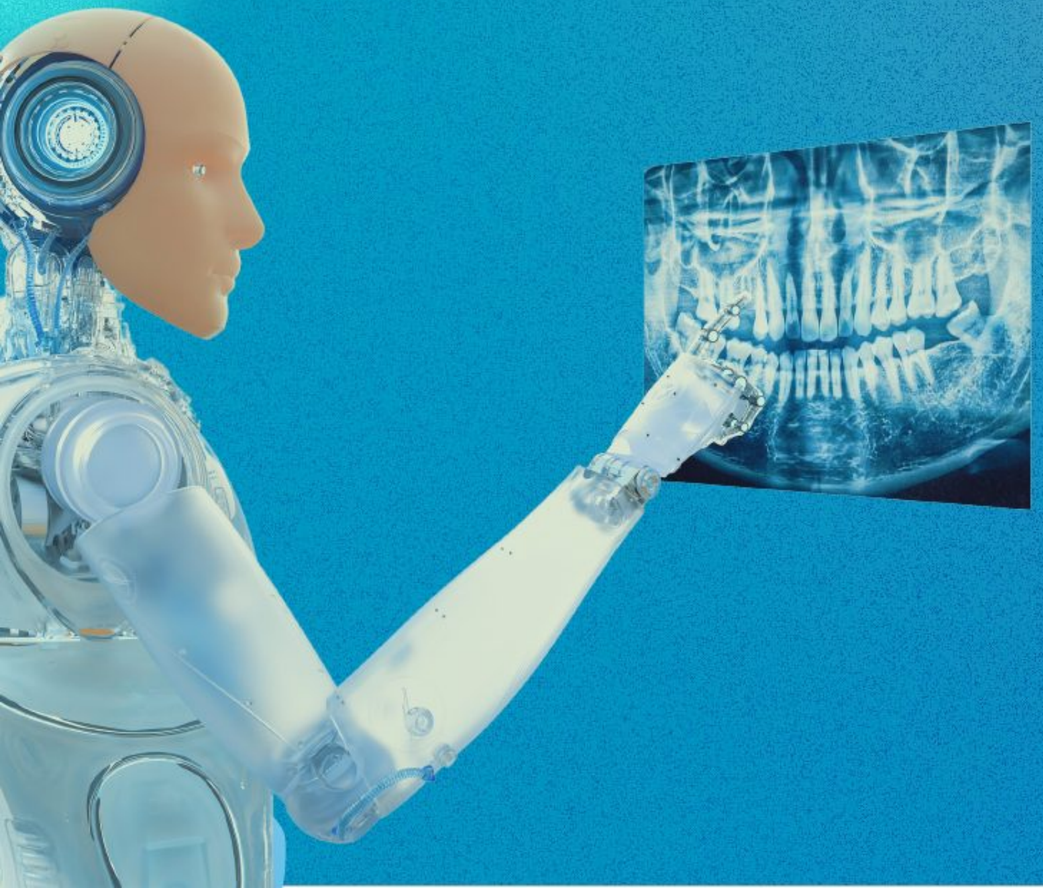
Rutgers AI rules vary by course and instructor, AI isn't permitted for coursework unless clearly stated, and students are responsible for following the rules

85% of physicians want a say in AI adoption, and students should expect one too

Ask, disclose, keep a human accountable

Only Rutgers-approved AI tools should be used for university work

Discussion Questions



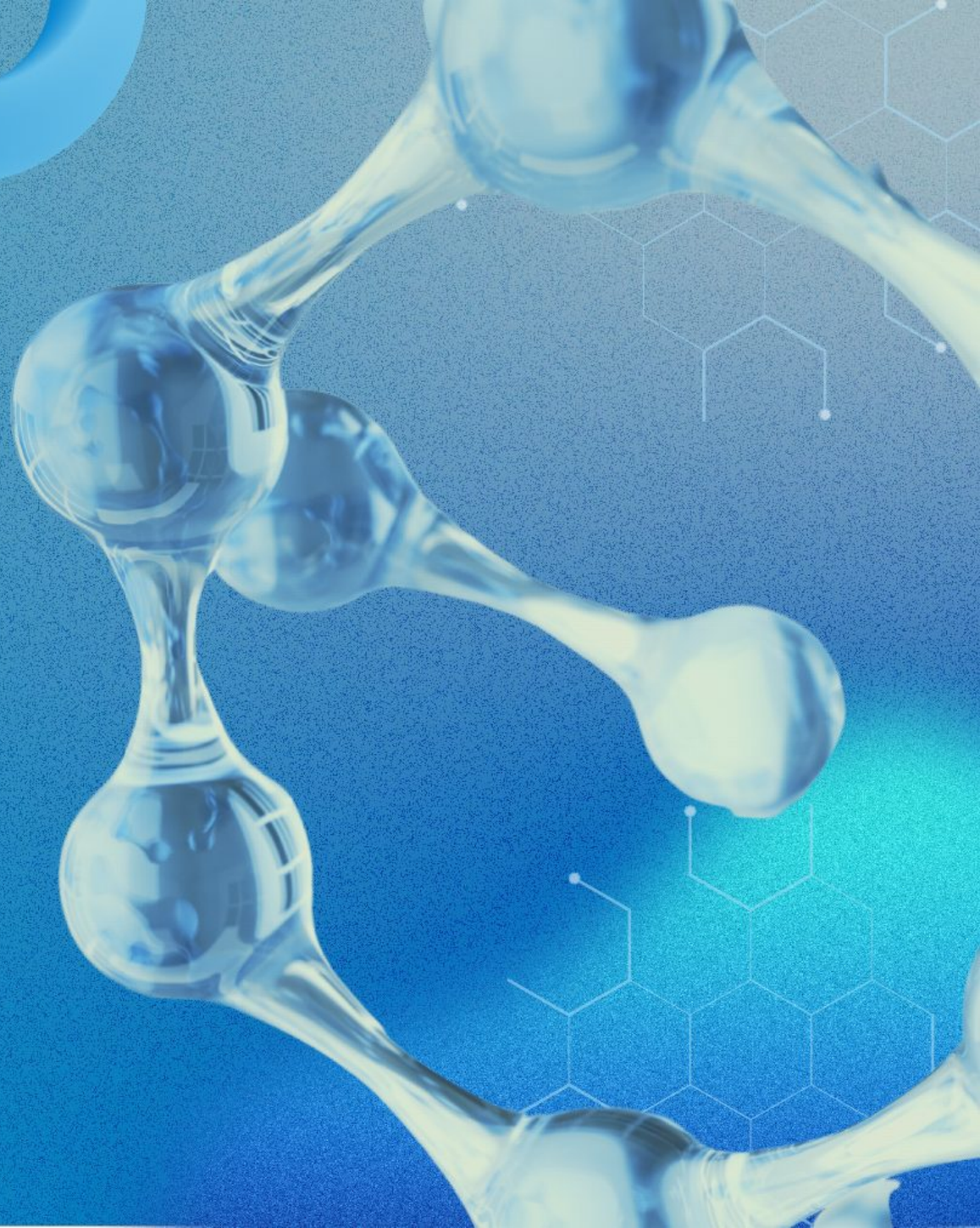
Should universities be required to tell students when AI is used in grading or advising?

Where is the line between using AI to learn and using it to avoid learning?

Who should be accountable when an AI tool gets something wrong, the student, the professor, or the university?

Key Takeaways

- AI is a tool, and its ethics depend on how we design and use it
- Transparency, accountability, and equity come up in every setting
- Use is outpacing training and policy
- Humans must stay in the loop



Acknowledgments

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Questions?



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